Supplementary Material: Leverage, Structural Rank, and Thermodynamic Selection in Information-Processing Systems

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
BA7	Physics.BoundedAcquisition.energy_ge_srank_cost paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	DQ1	DecisionQuotient.Physics.BoundedAcquisition.energy_ge_srank_cost paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
EP1	Physics.LocalityPhysics.landauer_principle	EP4	Physics.LocalityPhysics.nontrivial_physics paper4/DecisionQuotient/Physics/LocalityPhysics.lean
FXI1	fixed_axis_incompleteness paper1/axis_framework.lean	L1	matroid_basis_equicardinality paper1/axis_framework.lean
L2	Leverage.Physical.feasible_iff_floor_le_budget Leverage/Physical.lean	L3	Leverage.Physical.higher_leverage_same_caps_implies_lower_energy Leverage/Physical.lean
L4	l4_exchange_wrapper paper1/HandleAliases.lean	L5	l5_exchange_wrapper paper1/HandleAliases.lean
L6	nonorthogonal_complete_has_redundant_axis paper1/axis_framework.lean	L7	exists_semanticallyMinimal_subset paper1/axis_framework.lean
L8	exists_orthogonal_semanticallyMinimal_subset paper1/axis_framework.lean	L10	Leverage.SSOT.paper2_is_leverage_instance Leverage/SSOT.lean
L11	Leverage.SSOT.ssot_leverage_dominance Leverage/SSOT.lean	L13	Leverage.Typing.nominal_dominates_duck Leverage/Typing.lean
L14	Leverage.Typing.paper1_is_leverage_instance Leverage/Typing.lean	L15	Leverage.architecture_axes_independent Leverage/Probability.lean
L16	Leverage.bernoulli_justifies_linear_model Leverage/Probability.lean	L17	Leverage.compose_dof Leverage/Foundations.lean
L18	Leverage.composition_caps_additive Leverage/Theorems.lean	L19	Leverage.composition_dof_additive Leverage/Theorems.lean
L20	Leverage.correctness_probability Leverage/Probability.lean	L21	Leverage.dof_ratio_predicts_error_ratio Leverage/Probability.lean
L22	Leverage.dof_reliability_isomorphism Leverage/Probability.lean	L23	Leverage.error_independence_from_orthogonality Leverage/Probability.lean
L24	Leverage.error_probability_denom_pos Leverage/Probability.lean	L25	Leverage.expected_errors_from_linearity Leverage/Probability.lean
L26	Leverage.expected_errors_linear Leverage/Probability.lean	L27	Leverage.isomorphism_preserves_failure_ordering Leverage/Probability.lean
L28	Leverage.leverage_caps_principle Leverage/Theorems.lean	L29	Leverage.leverage_error_tradeoff Leverage/Theorems.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
L30	Leverage.leverage_gap Leverage/Probability.lean	L31	Leverage.leverage_maximization_principle Leverage/Theorems.lean
L32	Leverage.linear_model_preserves_ordering Leverage/Probability.lean	L33	Leverage.lower_dof_lower_errors Leverage/Probability.lean
L34	Leverage.max_leverage_is_optimal Leverage/Theorems.lean	L35	Leverage.metaprogramming_dominates Leverage/Theorems.lean
L36	Leverage.metaprogramming_unbounded_leverage Leverage/Theorems.lean	L37	Leverage.modification_eq_dof Leverage/Foundations.lean
L38	Leverage.optimal_minimizes_error Leverage/Theorems.lean	L39	Leverage.ordering_equivalence_exact Leverage/Probability.lean
L40	Leverage.series_error_probability Leverage/Probability.lean	L41	Leverage.system_is_correct Leverage/Probability.lean
L42	Leverage.testable_modification_prediction Leverage/Probability.lean	L43	dof_eq_srank Leverage/BridgeToDQ.lean
L44	dof_one_iff_max_leverage Leverage/Foundations.lean	L45	england_replication_inequality Leverage/BridgeToDQ.lean
L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean	L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean
L48	max_leverage_forces_dof_one Leverage/Foundations.lean	L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean
L50	ssot_max_leverage Leverage/Foundations.lean	L51	ssot_srank_one Leverage/BridgeToDQ.lean
L52	succ_le_two_pow Leverage/BridgeToDQ.lean	L53	sufficiency_conp_hard
L54	thermodynamic_selection Leverage/BridgeToDQ.lean	L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean
L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean	L57	Leverage.SSOT.modification_ratio Leverage/SSOT.lean
LP38	Physics.LocalityPhysics.pne_np_necessary_for_ physics paper4/DecisionQuotient/Physics/ LocalityPhysics.lean	LP43	Physics.LocalityPhysics.without_separation_no_ independence paper4/DecisionQuotient/Physics/ LocalityPhysics.lean
LP44	Physics.LocalityPhysics.without_finite_ capacity_no_gap paper4/DecisionQuotient/Physics/ LocalityPhysics.lean	ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 3.2: DOF = Independent Error Sources	L15, L23
Corollary 3.6: DOF-Error Monotonicity	L33
Corollary 3.13: DOF Ratio Predicts Error Ratio	L21, L42
Corollary 4.4: Leverage-Energy Monotonicity in a Capability Class	L3, L6
Corollary 3.5: Linear Approximation	L16, L39
Corollary 4.6: Physical Assumption Necessity Witnesses	L2, L8, L7, L4
Definition 2.4: Architecture	L17, L19
Theorem 3.10: Ordering Equivalence (Exact)	L32, L39
Theorem 4.9: Leverage Composition	L18, L19
Theorem 3.9: DOF-Reliability Isomorphism	L22, L27

Paper claim	**Lean handle**
Theorem 5.1: England Replication Inequality	L45
Theorem 3.3: Error Compounding	L20, L41
Theorem 3.1: Error Independence	L23
Theorem 3.4: Error Probability	L24, L40
Theorem 3.7: Expected Error Bound	L25, L26
Theorem 1.1: Five-Way Equivalence	*(no derived Lean handle found)*
Theorem 4.2: Leverage-Error Tradeoff	L29
Theorem 3.11: Leverage Gap	L30
Theorem 4.1: Leverage Maximization Principle	L28, L31
Theorem 4.7: Metaprogramming Dominance	L35, L36
Definition 2.12: Capability Set	L37
Unbounded Advantage	L13
Theorem 4.8: Optimal Architecture	L34, L38
Theorem 4.11: `AbstractClassSystem` as Leverage Instance	L13, L14
Theorem 4.12: `Ssot` as Leverage Instance	L10, L11, L13, L14
Theorem 4.5: Budget Feasibility Boundary	L2, L3, L4, L6
Theorem 4.3: Physical Edit-Energy Floor	L5, L1
SSOT Architecture	L11
Theorem 3.12: Testable Prediction	L30, L42

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.
Auto summary: mapped 28/29 (full=28, derived=0, unmapped=1).

3 Proof Hardness Index

Paper handle	**Hardness profile**	**Regime tags**	**Lean support**
`cor:dof-errors`	`unspecified`	-	L15, L23
`cor:dof-monotone`	`unspecified`	-	L33
`cor:dof-ratio`	`unspecified`	-	L21, L42
`cor:leverage-energy`	`unspecified`	-	L3, L6
`cor:linear-approx`	`unspecified`	-	L16, L39
`cor: physical-assumption-necessity`	`unspecified`	-	L2, L8, L7, L4
`prop:dof-additive`	`unspecified`	-	L17, L19
`thm:approx-bound`	`unspecified`	-	L32, L39
`thm:composition`	`unspecified`	-	L18, L19
`thm:dof-reliability`	`unspecified`	-	L22, L27
`thm:england`	`unspecified`	-	L45
`thm:error-compound`	`unspecified`	-	L20, L41
`thm: error-independence`	`unspecified`	-	L23

thm:error-prob	unspecified	-	L24, L40
thm:expected-errors	unspecified	-	L25, L26
thm:five-way	unspecified	-	*(no derived Lean handle found)*
thm:leverage-error	unspecified	-	L29
thm:leverage-gap	unspecified	-	L30
thm:leverage-max	unspecified	-	L28, L31
thm:metaprog	unspecified	-	L35, L36
thm:mod-bound	unspecified	-	L37
thm:nominal-leverage	unspecified	-	L13
thm:optimal	unspecified	-	L34, L38
thm: paper1-integration	unspecified	-	L13, L14
thm: paper2-integration	unspecified	-	L10, L11, L13, L14
thm: physical-budget-boundary	unspecified	-	L2, L3, L4, L6
thm: physical-energy-floor	unspecified	-	L5, L1
thm:ssot-leverage	unspecified	-	L11
thm: testable-prediction	unspecified	-	L30, L42

Auto summary: indexed 29 claims by hardness profile (unspecified=29).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- `Leverage/BridgeToDQ.lean`
- `Leverage/FiveWayEquivalence.lean`
- `Leverage/Physical.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)